

Alpha Diversity

Muluh

Community diversity

Group-wise comparisons

- **Index:** shannon
- **Group variable:** meal_group
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon

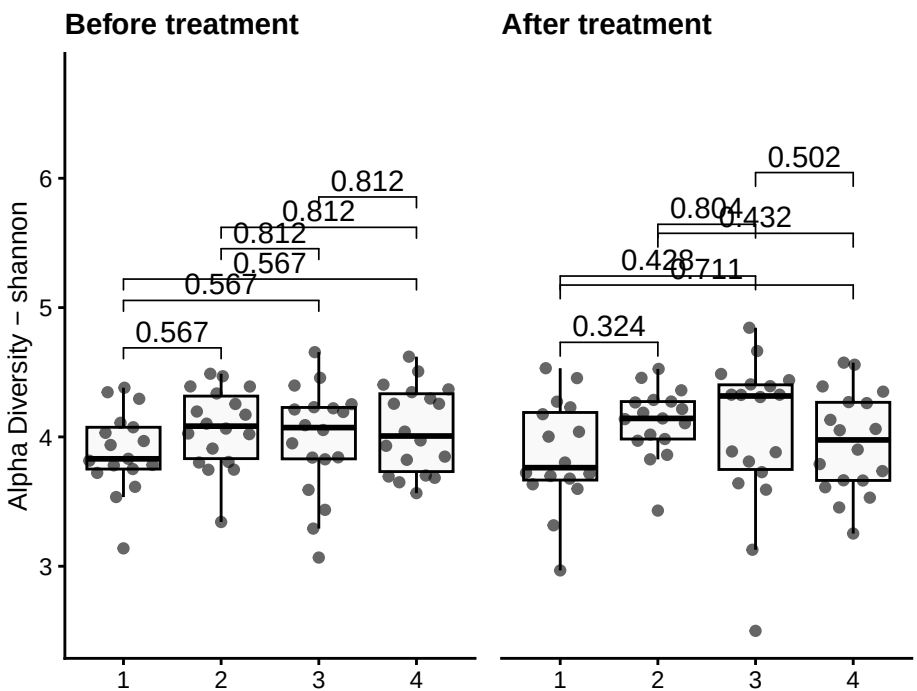


Table of the top findings

Across-group

Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

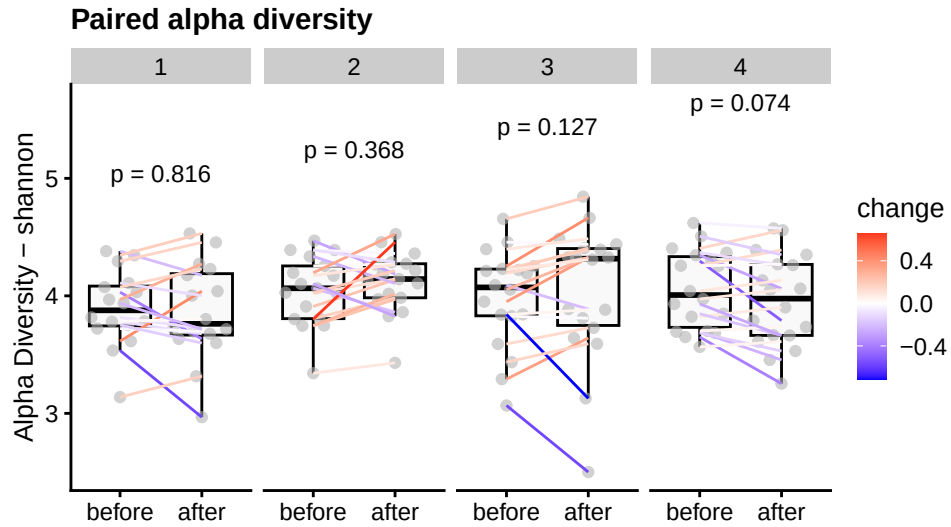
group1	group2	mean1	mean2	log2FC	p.adj
1	2	3.9	4.1	0.07	0.567
1	3	3.9	4.0	0.03	0.567
1	4	3.9	4.1	0.06	0.567
2	3	4.1	4.0	-0.03	0.812
2	4	4.1	4.1	-0.01	0.812
3	4	4.0	4.1	0.03	0.812

After treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	3.9	4.1	0.09	0.324
1	3	3.9	4.0	0.06	0.428
1	4	3.9	4.0	0.03	0.711
2	3	4.1	4.0	-0.03	0.804
2	4	4.1	4.0	-0.06	0.432
3	4	4.0	4.0	-0.03	0.502



Within-group

Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

meal_group	mean1	mean2	log2FC	p.value
1	3.9	3.9	-0.01	0.816
2	4.1	4.1	0.03	0.368
3	4.0	4.0	0.02	0.127
4	4.1	4.0	-0.03	0.074

Alternate analyses

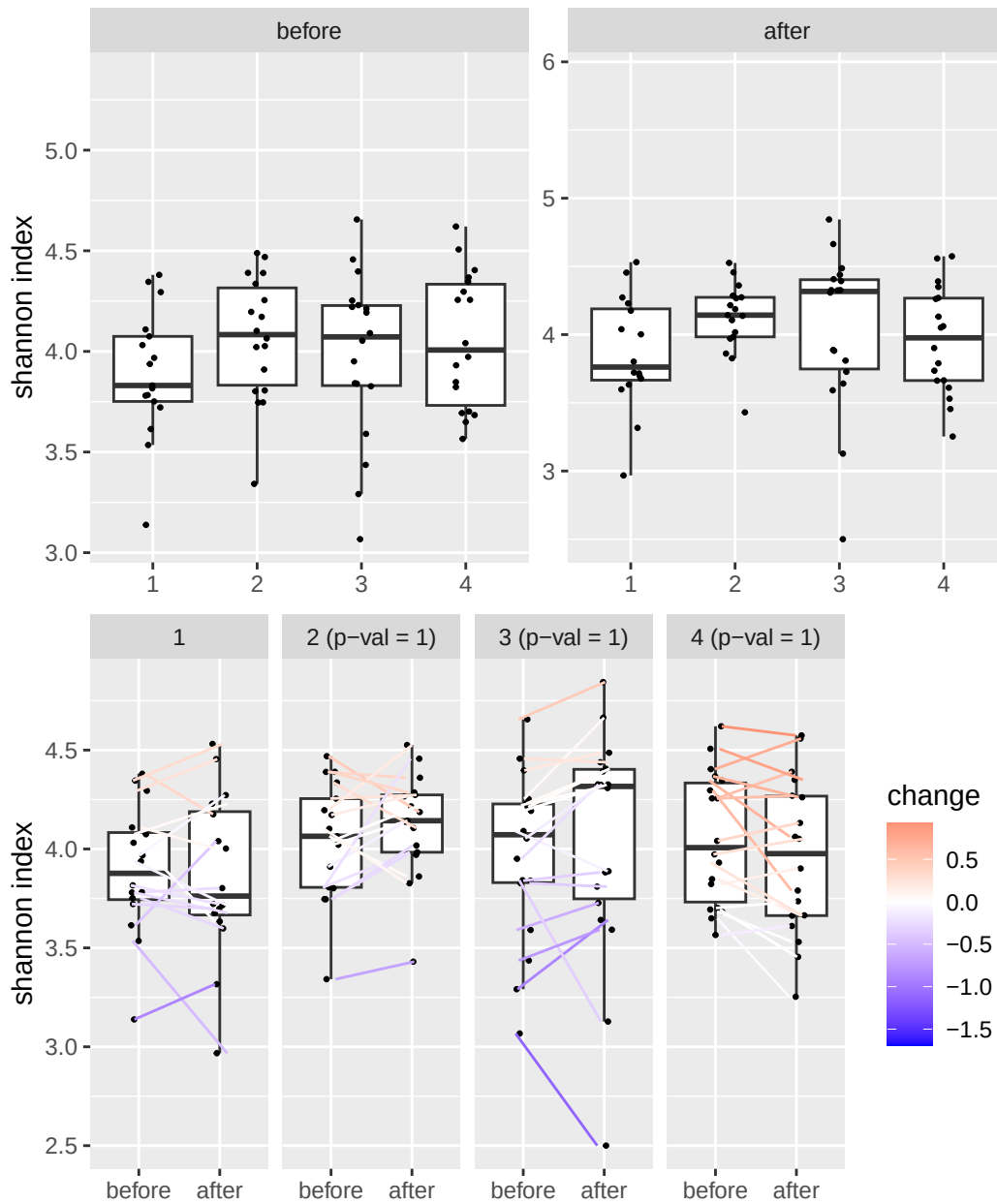
- **Beta (Estimate):** Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI:** Range where the true beta is likely to fall, reflecting precision.
- **p-Value:** Tests if beta differs from zero; $p < 0.05$ suggests significance.

Characteristic	before			after		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	3.6	3.1, 4.0	<0.001	3.5	2.9, 4.1	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.21	-0.03, 0.44	0.084	0.28	-0.02, 0.58	0.065
3	0.12	-0.12, 0.35	0.3	0.20	-0.09, 0.50	0.2
4	0.17	-0.06, 0.40	0.2	0.10	-0.20, 0.39	0.5
age	0.01	0.00, 0.01	0.2	0.01	0.00, 0.02	0.2

¹CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI ¹	p-value
(Intercept)	3.5	-Inf, Inf	<0.001
meal_group			0.4
1	—	—	
2	0.23	-Inf, Inf	>0.9
3	0.16	-Inf, Inf	>0.9
4	0.13	-Inf, Inf	>0.9
age	0.01	0.00, 0.01	0.039

¹CI = Confidence Interval



Data Summary

```

class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921

```

```
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```